SO2R Box — Proof-of-Concept Parts List

Bench-only POC. Every line is an Amazon search and is expected to be Prime-eligible from a major US warehouse (ELEGOO, SunFounder, Adafruit, HiLetgo, AZDelivery typically ship Prime). All wiring is on a breadboard — no soldering, no enclosure, no panel hardware. Skips the entire balanced audio chain, RS-232 transceivers, the ±5V rail, all 18 Omron RF relays, and every panel-mount connector.

Core POC subtotal: $83.00 · Stretch (audio mux + bring-up tools) adds $42.00 · Everything: **$125.00**.

What this POC proves

• **I2C control plane.** Heltec ESP32-S3 -> MCP23017 -> 8-ch relay module — the exact topology the RF box uses, scaled to 8 relays instead of 18.
• **Operator interface.** Four tactile buttons + four WS2812B LEDs + the Heltec's built-in OLED — renders status, accepts band/radio commands.
• **CAT-USB path.** One CP2102N module talking to a real radio over USB-C, validating the rig-control libraries before you order three of them.
• **Audio mux concept (stretch).** A single CD4051 fed from a phone's audio out into a headphone, with three GPIO lines selecting one of eight inputs — same chip the final design uses three of.
• **REVIEW B6 firsthand.** Drive the WS2812B chain directly from 3.3V GPIO and watch the colours go wrong — cheap, memorable proof that the level shifter is mandatory.

Core POC parts

Qty	Item	Search	Unit	Ext	Why you need it
1	Heltec WiFi LoRa 32 V3 (ESP32-S3, USB-C, built-in OLED)	Heltec WiFi LoRa 32 V3 ESP32-S3	$15.00	$15.00	The MCU. Same board the final build uses — POC firmware moves over unchanged.
1	Half-size solderless breadboard, 2-pack	half size breadboard 400 tie points 2 pack	$8.00	$8.00	One for the MCU + I2C bus, one for the audio mux experiment.
1	Dupont jumper wire kit (M-M, M-F, F-F, ~120 pcs)	dupont jumper wires kit male female 120 pieces	$8.00	$8.00	All inter-board wiring on the breadboard.
1	MCP23017 I2C 16-bit GPIO expander breakout	MCP23017 I2C breakout module	$8.00	$8.00	Validates the I2C control plane. Same chip the RF box uses to drive 18 relays — POC drives 8.
1	8-channel 5V relay module (opto-isolated)	8 channel 5V relay module opto isolated	$10.00	$10.00	Stand-in for the 18x Omron G2RL. Coils are 5V (powered from USB 5V or a 5V USB charger - NOT from the 12V adapter below). Opto inputs accept the Heltec's 3.3V GPIO directly.
1	Tactile push-button assortment (12mm, breadboard-friendly)	tactile push button 12mm breadboard assortment	$6.00	$6.00	Four buttons = the four panel buttons (Radio Select x2, Band -, Band +).
1	WS2812B 5mm through-hole RGB LED, 10-pack	WS2812B 5mm through hole RGB LED 10 pack	$7.00	$7.00	Four LEDs = panel status. Confirms the level-shifter requirement (REVIEW B6) for yourself.
1	CP2102N USB-C to UART module	CP2102N USB-C UART bridge module	$7.00	$7.00	Validates one CAT-USB path to a real radio. Buy one, prove the path, then order the other two.
1	12V / 1A wall adapter, 2.1mm center-positive	12V 1A power supply 2.1mm center positive	$8.00	$8.00	Reserved for the final build (the real G2RL relays are 12V coil). NOT used for the POC - it would burn the 5V module's coils. Roll forward to the real RF box.
1	2.1mm DC barrel jack to screw-terminal adapter, 5-pack	2.1mm DC barrel jack screw terminal adapter	$6.00	$6.00	Lets the wall adapter plug into a breadboard rail. Easier than chopping the cable.
		Subtotal		**$83.00**	

Stretch parts (audio mux + bring-up tools)

Qty	Item	Search	Unit	Ext	Why you need it
1	CD4051BE 8:1 analog mux, DIP-16, 5-pack	CD4051BE DIP16 5 pack	$7.00	$7.00	Audio routing POC. Three address pins from the ESP32 select 1 of 8 inputs.
1	Resistor assortment, 1/4W metal-film, 525 piece	1/4W metal film resistor assortment kit 525	$7.00	$7.00	Pull-ups, current-limit on LEDs, audio bias. Covers every value the final BOM needs.
1	Ceramic capacitor assortment (>=100nF qty 100)	ceramic capacitor assortment kit 0.1uF	$8.00	$8.00	100nF IC bypass on every chip — bring up the I2C bus cleanly the first time.
1	3.5mm TRS panel jack to bare wire, breakout 5-pack	3.5mm audio jack breakout module 5 pack	$8.00	$8.00	Patch a phone's audio into the CD4051 input + a headphone into the output to hear the mux switch.
1	USB logic analyzer, 8-channel, 24 MHz (Saleae-clone)	USB logic analyzer 8 channel 24MHz	$12.00	$12.00	Pulseview / Sigrok compatible. Indispensable for I2C and WS2812B bring-up. Skip if you already own one.
		Subtotal		**$42.00**	

Not included on purpose

• **Omron G2RL relays ($82).** The relay module above is functionally identical for logic-validation; you only need the real Omrons when you wire actual RF.
• **THAT1646 + INA333 + TC7660H mic chain (~$30).** Audio in/out can be POCed with a phone and headphones into the CD4051 — the balanced chain is a separate POC.
• **MAX3232 RS-232 quartet (~$9).** Use USB-CAT (CP2102N) against a USB-capable radio for POC; RS-232 is just a level translation that adds nothing architecturally.
• **SC16IS752 UART expanders (~$14).** Heltec native UARTs + USB CDC give you three CAT paths for free — enough to prove SO2R control of two radios.
• **P82B96 + Cat5e inter-box link (~$5).** Run everything on one breadboard; the two-box split is a packaging decision, not an architectural one.
• **BPF-8A modification parts (~$10).** Defer until you've validated the band-switch command pipeline against the 8-ch relay module.
• **All panel-mount connectors, the enclosure, the PCB.** Bench POC. Breadboard only.

Suggested POC milestones (4 evenings)

1. Blink + OLED + button (1 hr). Flash the Heltec, light an LED on the integrated OLED, read a button. Confirms the toolchain.
2. I2C bring-up (1-2 hrs). Probe MCP23017 from the ESP32, click each relay on the 8-ch module in sequence. Verify ULN2803 isn't needed for the POC (the relay module has its own driver transistors).
3. Panel UI (1 evening). Four buttons -> band/radio state machine. Four WS2812B LEDs reflect state. OLED shows current band + active radio.
4. CAT control (1 evening). CP2102N to a real radio's USB-CAT port. Implement frequency/band read + PTT through your chosen rig-control library. *This is the riskiest leg of the build — do it before committing to the rest of the BOM.*
5. (Stretch) Audio mux (1 evening). CD4051 between a phone and a headphone. Address lines from three GPIOs. Listen for the click-through on each channel select.

Prime/shipping notes

Search URLs above intentionally show multiple sellers per item so you can pick the fastest-arriving Prime listing on the day. Counterfeit risk is real for the CP2102N and MCP23017 — if the module behaves strangely on the bench, swap to an Adafruit-branded listing (same Amazon site, costs ~2x) rather than re-debugging your firmware. Total POC weight is well under 1kg; a single Amazon Prime order usually consolidates to 1-2 deliveries.